Investigating the Robustness of Non-negative Matrix Factorisation Algorithms towards Outliers

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Abstract

Non-negative matrix factorisation (NMF) is widely used to learn interpretable representations of face images, yet real images are often partly occluded, and it is unclear how much the choice of loss function affects robustness to such corruption. In this report we compare two formulations that differ only in how they penalise reconstruction error. Standard NMF minimises the squared Frobenius norm, so every residual is squared and a few heavily occluded images can dominate the fit, whereas $L_{2,1}$-norm NMF sums the error of each image without squaring it, which gives heavily corrupted images a smaller weight. We evaluate both on the ORL and Extended YaleB face datasets, varying the number, size and spatial extent of the occlusion, the fraction of corrupted images and the factorisation rank, and compare them by the relative reconstruction error (RRE) against the clean images and by clustering accuracy and normalised mutual information. We find that $L_{2,1}$-NMF is clearly more robust when only a minority of images is corrupted: with 10% of the ORL images carrying 350 occluded pixels each, its RRE is about 0.16 against 0.23 for standard NMF and its clustering accuracy about 0.68 against 0.41, and the gap widens as the rank grows. Its advantage shrinks when the corruption forms one large contiguous block, disappears as the fraction of corrupted images approaches one, and is small when the learned basis is transferred to unseen images. These comparisons show that $L_{2,1}$-NMF suits datasets with a few heavily corrupted samples, while dataset-wide or large structured occlusion calls for other approaches.

1 Introduction

Non-negative matrix factorisation (NMF) approximates a non-negative data matrix by the product of two smaller non-negative factors. Because the factors cannot cancel each other out, each image is rebuilt by adding parts together, and on face data the learned basis images resemble eyes, noses and mouths rather than the global, signed patterns produced by principal component analysis [1]. This interpretability, together with simple multiplicative update rules [2], has made NMF a standard tool for dimensionality reduction, feature learning and clustering of image data [3].

Real face images, however, are rarely clean. Sunglasses, scarves, hands, hair and strong shadows hide part of the face, and the hidden pixels bear no relation to the identity underneath. Standard NMF is poorly suited to this situation. It minimises the squared Frobenius norm of the reconstruction error, which is optimal only when the noise is small and Gaussian. An occluded pixel produces a residual

far larger than that, and squaring it lets a small number of corrupted entries dominate the objective. The model then spends its limited capacity fitting the occluders, its basis images absorb bright patches, and both the reconstruction of the true faces and any downstream task built on the learned representation, such as clustering by identity, deteriorate. Since occlusion is common in practice and NMF is often applied without any cleaning step, understanding how to make the factorisation robust to it is a problem of direct practical importance.

Many robust variants have been proposed. Some replace the squared loss by a less sensitive one, such as the elementwise ℓ_1 norm [4], the sample-wise $L_{2,1}$ norm [5] or the hypersurface cost [6]; others model the corruption explicitly with a sparse error term [7] or truncate large errors [8]. Each was, however, introduced under its own noise model and evaluated on its own data, so the literature does not say when a robust loss actually helps. Does the benefit hold for scattered occlusion as well as for one large block? Does it survive when most images are corrupted rather than a few? Does a more robust basis also transfer better to unseen images? A previous comparison on the same face datasets could not answer these questions, because its iteration budget was too small for the robust methods to converge [9]. Without such answers, a practitioner has no principled way to decide whether replacing standard NMF is worth it for a given dataset.

This report addresses that gap with a controlled comparison that isolates the effect of the loss function. We compare standard NMF with $L_{2,1}$-norm NMF, which differ only in how the reconstruction error is penalised: the former squares the error of every pixel, while the latter measures the error of each image as a whole and sums these errors without squaring them. The intuition is that a heavily occluded image then contributes in proportion to its error rather than its squared error, and the resulting algorithm automatically assigns such images a smaller weight. $L_{2,1}$-NMF is also a natural candidate in practice: it keeps the same multiplicative updates as the baseline, adds almost no computational cost and introduces no extra hyperparameter. Both algorithms are implemented from scratch and share the same initialisation, random seeds and iteration budget, so that any difference in the results can be attributed to the loss alone.

We evaluate the two methods on the ORL and Extended YaleB face datasets under synthetic occlusion, varying the number of occluded regions, their size and spatial extent, the fraction of corrupted images and the factorisation rank, and we also test how well a basis learned from corrupted data generalises to unseen clean images. Performance is measured by the relative reconstruction error against the clean images and by clustering accuracy and normalised mutual information. The results show that $L_{2,1}$-NMF is clearly more robust when a minority of images is corrupted, with the gap widening as the rank grows, but that its advantage shrinks for large contiguous blocks, disappears when every image is corrupted, and is small when the basis is transferred to new images.

The remainder of this report is organised as follows. Section 2 reviews related work on NMF and its robust variants. Section 3 describes the two algorithms and the occlusion model. Section 4 presents the experimental setup and results, and Section 5 concludes with limitations and directions for future work.

2 Related work

Matrix factorisation methods approximate a data matrix by a product of two low rank factors, and thereby expose a small set of components that explain the data. Principal component analysis and the singular value decomposition allow the entries of both factors to take either sign, so a component may be subtracted from another; the resulting basis vectors are hard to read as objects in their own right. Non-negative matrix factorization (NMF) forbids negative entries, which restricts the reconstruction to purely additive combinations and makes the learned components interpretable as parts of the data. A recent survey by Saberi-Movahed et al. [3], covering the literature from 2000 to 2025, identifies robust NMF as one of the main research directions of the field, which reflects how central the sensitivity to corrupted data has become. This section reviews the classical formulation, the robust variants that modify its loss function, and the variants that model the corruption explicitly.

2.1 Classical NMF

Lee and Seung [1] popularised NMF by showing that the non-negativity constraint alone is enough to recover a parts-based representation of face images, with basis vectors that correspond to localised

facial features. The companion paper [2] gives the multiplicative update rules that remain the standard solver: each rule is a gradient step with an elementwise step size chosen so that the iterates stay non-negative, and the objective is shown to be non-increasing under them for both the squared Frobenius loss and the generalised Kullback-Leibler divergence. The appeal of these rules is that they need no step size tuning, no projection, and only a few matrix products per iteration.

The formulation has known weaknesses. The factorisation is not unique and the objective is not jointly convex, so different initialisations yield different solutions. Convergence is slow near a solution, and a non-increasing objective does not by itself imply convergence to a stationary point; Lin [10] analyses this gap and proposes modified updates that close it. Most importantly for this report, the squared Frobenius loss corresponds to a Gaussian noise model, which makes the estimator fragile when a small number of entries are grossly corrupted.

The multiplicative rules are not the only option. Treating the problem as a sequence of constrained subproblems leads to projected gradient methods [11], to alternating non-negative least squares, and to hierarchical alternating least squares, which updates one rank-one component at a time; the recent survey of Saberi-Movahed et al. [3] reviews these families. They typically reach a lower objective in fewer iterations and come with stronger stationarity guarantees, at the price of an inner solver, a line search, or a step size to tune. Because this report compares cost functions rather than solvers, we keep the multiplicative rules for every algorithm, so that any difference in robustness can be attributed to the loss and not to the optimiser.

2.2 Robust NMF through the loss function

The most direct way to limit the influence of a corrupted entry is to replace the squared residual by its absolute value. Lam [4] derives such an algorithm from a Laplacian rather than Gaussian or Poisson noise model, formulates the factorisation as an ℓ_1 minimisation and solves it by linear programming. The elementwise penalty grows linearly, so no single pixel can dominate the objective, and the model is a natural fit for heavy-tailed corruption. The difficulty is computational: the absolute value is not differentiable at the origin, so the simple multiplicative scheme no longer applies, and Guan et al. [12] report that Lam's solver converges slowly and does not scale, which motivates their own first-order method for the same objective. Seraghiti et al. [13] sharpen the point by proving that ℓ_1-NMF is NP-hard even for rank one, in contrast to the least squares formulation. A further practical cost is that any reweighting scheme for this loss carries a weight for every entry, a matrix as large as the data itself.

Kong et al. [5] aggregate the residual differently: the errors of one data point are first combined by an ℓ_2 norm and the resulting per-sample errors are then summed without squaring, which yields the $\ell_{2,1}$ norm of the residual matrix. A sample with a large error therefore contributes linearly rather than quadratically, and the resulting algorithm can be written as a reweighted version of the classical updates in which each sample carries a single scalar weight. This keeps the solver almost as cheap as the baseline, which is why we adopt it here; the cost function and its derivation are given in Section 3.2. The formulation does assume that corruption is concentrated in a subset of samples, since down-weighting a sample discards its clean features along with its corrupted ones.

Hamza and Brady [6] take a third route and replace the quadratic penalty by the hypersurface cost $\varphi(t) = \sqrt{1 + t^2} - 1$, which grows quadratically for small residuals and linearly for large ones. The function is smooth everywhere, so the resulting updates are as simple as the classical ones and no reweighting is needed. The price is that the loss sits between the quadratic and the linear regime, so the gain in robustness is smaller than that of an explicitly robust norm, and the original evaluation concerns reflectance spectra rather than images with structured occlusion.

2.3 Robust NMF through explicit noise modelling

Zhang et al. [7] observe that the classical objective is optimal under Gaussian or Poisson noise and therefore unsuitable when the data are grossly corrupted. Their robust NMF decomposes the data matrix as the sum of a sparse error matrix and a product of two non-negative factors, so that arbitrarily corrupted entries are absorbed by the sparse term instead of distorting the factors, and they validate the method on two face databases. The assumption matches contiguous occlusion well,

Table 1: Summary of the reviewed formulations. $R = X - WH$ denotes the residual, $R_{\cdot j}$ its j-th column, and S a sparse corruption matrix.

Method	Loss on the residual	Advantage	Limitation
Standard NMF [1, 2]	$\|R\|_F^2$	simple updates, no step size, parts-based bases	Gaussian assumption, a few extreme entries dominate
L1-norm NMF [4]	$\|R\|_1$	linear penalty on every entry, fits heavy-tailed noise	non-differentiable, NP-hard, entrywise weights
L2,1-norm NMF [5]	$\sum_j \|R_{\cdot j}\|_2$	one scalar weight per sample, cost close to the baseline	discards the clean part of a down-weighted sample
Hypersurface cost [6]	$\sum_{ij}(\sqrt{1 + R_{ij}^2} - 1)$	smooth everywhere, quadratic near zero and linear for large errors	robustness gain smaller than an explicitly robust norm
Robust NMF [7]	$\|X - WH - S\|_F^2 + \lambda\|S\|_1$	models the corruption explicitly, S can be inspected	extra parameter λ the solution is sensitive to

since a block is both sparse in area and extreme in value. The cost is an additional regularisation parameter that controls the sparsity of the error term and to which the solution is sensitive.

Dai et al. [14] attack the same problem from the opposite end: instead of estimating the corruption, they build a binary weight matrix that labels each entry as clean or corrupted and factorise only the clean part, reporting results on images corrupted by salt-and-pepper noise and by contiguous occlusion. The formulation is simple and directly targeted at the noise model used in this report, but it depends on how reliably the corrupted entries can be identified in the first place.

Guan et al. [8] push the same idea further by truncating large errors through a Truncated Cauchy loss, and provide a generalisation bound as well as an empirical study on corrupted image data. Their paper also surveys the robust NMF literature and is a useful entry point to it.

Table 1 summarises the formulations reviewed above.

2.4 Positioning of this report

The formulations above were each introduced under a different corruption model, from Laplace noise on image features to sparse gross errors, and were evaluated on different data, so their relative behaviour under one common protocol cannot be read off the literature. Where the same two algorithms have been compared on these datasets, the comparison has been limited by computation: Yang et al. [9] cap the iterations so tightly that their robust methods never reach convergence, which by their own account leaves the robustness question open. This report therefore implements the classical baseline and $\ell_{2,1}$-norm NMF from scratch, keeps the solver family, the initialisation and the random seeds identical across algorithms, gives both the same and much larger iteration budget, and compares them on ORL and Extended YaleB under occlusion of varying number, size and spatial extent, applied to a varying fraction of the images.

3 Methods

3.1 Standard NMF

3.1.1 Model setup

Let

$$X = [x_1, x_2, \ldots, x_n] \in \mathbb{R}_+^{m \times n} \tag{1}$$

denote the non-negative data matrix, where each column x_i is one vectorised face image, m is the number of pixels and n the number of images. NMF seeks two non-negative factors

$$W \in \mathbb{R}_+^{m \times r}, \qquad H \in \mathbb{R}_+^{r \times n}, \qquad r \ll \min(m, n), \tag{2}$$

such that $X \approx WH$. Each column of W is a basis image and the i-th column h_i of H holds the non-negative coefficients that reconstruct x_i as an additive combination of those basis images. Throughout, $\odot$ denotes elementwise multiplication, division between matrices is elementwise, and $R = X - WH$ is the residual matrix.

3.1.2 Objective Function and Loss Formulation

The baseline follows Lee and Seung [1, 2] and minimises the squared Frobenius norm of the residual:

$$\min_{W \geq 0,\, H \geq 0} J_F(W, H) = \|X - WH\|_F^2 = \sum_{i=1}^{n} \|x_i - Wh_i\|_2^2. \tag{3}$$

This choice has a precise statistical meaning. If every entry is generated as $X_{ji} = (WH)_{ji} + \varepsilon_{ji}$ with independent Gaussian noise $\varepsilon_{ji} \sim \mathcal{N}(0, \sigma^2)$, the negative log likelihood of X equals $J_F/(2\sigma^2)$ up to an additive constant, so minimising (3) is maximum likelihood estimation under an i.i.d. Gaussian noise model. The method therefore inherits both the convenience and the fragility of that assumption, as discussed in Section 3.1.6.

Problem (3) is not jointly convex in (W, H), but it is convex in each factor when the other is fixed, which motivates alternating updates.

3.1.3 Multiplicative Update Rules

The gradient with respect to W is $\nabla_W J_F = 2(WHH^\top - XH^\top)$. A plain gradient step $W \leftarrow W - \eta \odot \nabla_W J_F$ would need a projection to keep W non-negative. Lee and Seung [2] instead choose the step size elementwise as $\eta = W/(2WHH^\top)$, and the additive step collapses into a multiplicative one:

$$W^{(t+1)} = W^{(t)} \odot \frac{X(H^{(t)})^\top}{W^{(t)}H^{(t)}(H^{(t)})^\top + \delta}. \tag{4}$$

The symmetric argument for H gives

$$H^{(t+1)} = H^{(t)} \odot \frac{(W^{(t+1)})^\top X}{(W^{(t+1)})^\top W^{(t+1)}H^{(t)} + \delta}. \tag{5}$$

As in Section 3.2, $\delta > 0$ is a small constant added to the denominators for numerical stability, and W is updated before H so that both algorithms follow the same alternating order.

In each rule the numerator is the negative part of the gradient and the denominator the positive part, which gives the scheme three properties. First, non-negativity is preserved automatically, so no projection is required provided the factors are initialised with positive entries. Second, where the gradient vanishes the numerator and the denominator coincide, the ratio equals one, and the factors stop changing, so stationary points are fixed points of the iteration. Third, the objective is non-increasing: Lee and Seung establish this with an auxiliary function $G(h, h^{(t)})$ satisfying $G(h, h^{(t)}) \geq F(h)$ and $G(h, h) = F(h)$, whose minimiser recovers (5) and yields $F(h^{(t+1)}) \leq G(h^{(t+1)}, h^{(t)}) \leq G(h^{(t)}, h^{(t)}) = F(h^{(t)})$. A monotone objective does not by itself guarantee convergence to a stationary point; Lin [10] analyses this gap and proposes modified updates that close it. In practice the plain rules behave reliably on image data, which is why they remain the standard baseline and the natural reference point for the robust variant.

3.1.4 Implementation procedure

The complete optimisation procedure is summarised as follows.

1. Initialise W and H with non-negative random values.
2. Update W using Equation (4).
3. Update H using Equation (5).
4. Compute the current objective value $J_F^{(t)} = \sum_{i=1}^{n} \|x_i - W^{(t)}h_i^{(t)}\|_2^2$.

5. Repeat Steps 2 to 4 until the stopping criterion is satisfied or the maximum number of iterations is reached.

The initialisation is shared with L2,1-NMF and described in Section 3.2.5, so both algorithms start from identical random factors within a run, and both receive the same iteration budget and tolerance, given in Section 4. The baseline is not cheap to converge: on the clean ORL data it needs roughly 2300 iterations to reach a relative change of 10^{-5} (Table 2), so under the shared budget it may stop at the iteration cap rather than at the tolerance. Because the cap is identical for the two algorithms, the comparison is made at equal computational budget. The matrix products are grouped so that the small $r \times r$ Gram matrices $HH^\top$ and $W^\top W$ are formed first; evaluating $W^\top(WH)$ instead would build an additional $m \times n$ intermediate at every iteration.

3.1.5 Computational complexity

The cost of one iteration is dominated by the matrix products in (4) and (5), giving an approximate per-iteration complexity of

$$O(mnr), \tag{6}$$

the same leading order as L2,1-NMF. Standard NMF avoids the additional residual norm and weight computation of the reweighted scheme, so its constant factor is slightly smaller.

3.1.6 Sensitivity to occlusion noise

The Gaussian assumption behind (3) is badly violated by the block occlusion described in Section 3.3. An occluded pixel is set to the maximum intensity irrespective of the face underneath, so its residual is far larger than anything a Gaussian with small σ would plausibly produce. Three consequences follow.

First, the penalty is quadratic. A residual of magnitude ρ contributes ρ^2 to J_F, so a pixel whose residual is four times the typical one contributes sixteen times as much, and a small number of occluded pixels can outweigh thousands of well reconstructed ones. Second, the influence of an entry on the updates is unbounded. Writing the gradient as $\nabla_W J_F = -2RH^\top$ shows that every pixel pulls on the basis images in proportion to its residual, with no upper limit and no mechanism for recognising an entry as an outlier; the multiplicative rules inherit exactly this weighting through the terms $XH^\top$ and $W^\top X$. Third, model capacity is wasted. Because the blocks appear at a different position in every image, a rank-r factorisation cannot represent them compactly, yet minimising (3) forces the model to try, so basis images acquire bright patches and the reconstruction of genuine face pixels deteriorates.

Table 2 makes these effects concrete on ORL. For each block size we ran Standard NMF with $r = 40$ and split the final residual into occluded and non-occluded pixels. With $b = 3$, fewer than one percent of the entries are occluded, yet they account for a quarter of the objective, an over-representation of more than thirty times. As b grows, the residual on occluded pixels falls while the residual on genuine face pixels rises by a third relative to the clean fit, which is the wasted capacity effect described above. Increasing the number of blocks acts in the same direction: at $b = 6$, the share of the objective attributable to occluded pixels grows from 19.2% with one block to 24.9% with two and 28.2% with three. This is precisely the failure mode that the L2,1 formulation of Section 3.2 is designed to limit.

A block of a given edge length covers different fractions of the two datasets, 3.2% of a 30×37 ORL image against 2.0% of a 42×48 Extended YaleB image for $b = 6$, which is why Section 3.3 uses a separate set of block sizes for each dataset and why block size and occlusion fraction must be distinguished when results are compared across datasets.

3.2 $L_{2,1}$-Norm Based Non-negative Matrix Factorization

3.2.1 Motivation

Standard Non-negative Matrix Factorization (NMF) commonly minimises the squared Frobenius norm between the observed data matrix and its low-rank approximation. However, the squared loss assigns disproportionately large penalties to samples with large reconstruction errors. Consequently,

Table 2: Diagnostic of Standard NMF on ORL (30×37 pixels, $r = 40$, one occlusion block of edge b per image, seed 42). Every run was iterated until the relative change of the objective fell below 10^{-5}; the iteration count required is reported in the last column. RMS residuals are computed separately over occluded and non-occluded pixels.

b	occluded pixels	share of J_F	RMS (occluded)	RMS (face)	iterations
0	0.0%	0.0%	–	0.060	2339
3	0.8%	25.1%	0.412	0.064	2165
6	3.2%	19.2%	0.205	0.077	1923
9	7.3%	22.9%	0.156	0.080	2009

a small number of heavily corrupted samples (large noise or sometimes called outliers) may substantially affect the learned basis matrix.

To reduce the influence of such samples, we employ $L_{2,1}$-norm based NMF ($L_{2,1}$-NMF). Instead of squaring the reconstruction error of each sample, $L_{2,1}$-NMF sums the Euclidean norms of the sample-wise residual vectors. Therefore, samples with unusually large reconstruction errors have a less dominant influence on the factorisation process.

3.2.2 Objective Function and Loss Formulation

Non-negative matrix factorisation (NMF) represents a non-negative data matrix as the product of two lower-rank non-negative matrices, producing an additive and often parts-based representation of the original data [1]. Let

$$X = [x_1, x_2, \ldots, x_n] \in \mathbb{R}_+^{m \times n} \tag{7}$$

denote the non-negative input matrix, where each column x_i represents one vectorised face image, m is the number of pixels, and n is the number of images. The objective is to find two non-negative low-rank matrices

$$W \in \mathbb{R}_+^{m \times r}, \qquad H \in \mathbb{R}_+^{r \times n}, \tag{8}$$

such that

$$X \approx WH, \tag{9}$$

where r is the factorisation rank.

The standard NMF objective based on the squared Frobenius norm is

$$\min_{W,H \geq 0} \|X - WH\|_F^2 = \min_{W,H \geq 0} \sum_{i=1}^{n} \|x_i - Wh_i\|_2^2, \tag{10}$$

where h_i is the i-th column of H [2].

To reduce the influence of noisy samples and outliers, Kong et al. [5] proposed a robust NMF formulation based on the $\ell_{2,1}$-norm:

$$\min_{W,H \geq 0} \|X - WH\|_{2,1}, \tag{11}$$

where the $L_{2,1}$ norm is defined as

$$\|X - WH\|_{2,1} = \sum_{i=1}^{n} \|x_i - Wh_i\|_2. \tag{12}$$

[5]

The inner L_2 norm measures the reconstruction error of each individual image, while the outer L_1 aggregation sums these sample-wise errors without squaring them. As a result, heavily corrupted samples receive less influence than under the squared Frobenius loss.

3.2.3 Iteratively Reweighted Update Rules

The objective in Equation (11) is non-smooth when a sample has zero reconstruction error. Following Kong et al. [5], we adopt an iteratively reweighted optimisation strategy.

At iteration t, the residual matrix is calculated as

$$R^{(t)} = X - W^{(t)}H^{(t)}, \tag{13}$$

where the i-th column of $R^{(t)}$ is

$$r_i^{(t)} = x_i - W^{(t)}h_i^{(t)}. \tag{14}$$

A diagonal weighting matrix is then constructed as

$$D^{(t)} = \operatorname{diag}\left(d_1^{(t)}, d_2^{(t)}, \ldots, d_n^{(t)}\right), \tag{15}$$

where

$$d_i^{(t)} = \frac{1}{2\sqrt{\|r_i^{(t)}\|_2^2 + \epsilon}}. \tag{16}$$

Here, $\epsilon > 0$ is a small numerical constant used to avoid division by zero. In our implementation, ϵ is set to 10^{-10}.

Equation (16) assigns a smaller weight to a sample with a larger reconstruction error. Therefore, a severely corrupted image has less influence on the learned basis matrix and the corresponding low-dimensional subspace.

With $D^{(t)}$ fixed, the non-smooth $L_{2,1}$ objective is approximated by the following weighted quadratic surrogate:

$$Q(W, H \mid W^{(t)}, H^{(t)}) = \operatorname{Tr}\left[(X - WH)D^{(t)}(X - WH)^\top\right] + C^{(t)}, \tag{17}$$

where $C^{(t)}$ is constant with respect to W and H during the current iteration. Therefore, it does not affect the updates of the two factor matrices.

The factor matrices are updated using the following weighted multiplicative rules:

$$W^{(t+1)} = W^{(t)} \odot \frac{XD^{(t)}(H^{(t)})^\top}{W^{(t)}H^{(t)}D^{(t)}(H^{(t)})^\top + \delta}, \tag{18}$$

and

$$H^{(t+1)} = H^{(t)} \odot \frac{(W^{(t+1)})^\top XD^{(t)}}{(W^{(t+1)})^\top W^{(t+1)}H^{(t)}D^{(t)} + \delta}. \tag{19}$$

In Equations (18) and (19), $\odot$ denotes element-wise multiplication, and the division is also performed element-wise. The small constant $\delta > 0$ is added to the denominators to improve numerical stability.

Since all entries of X, W, H, and $D^{(t)}$ are non-negative, the multiplicative updates preserve the non-negativity of W and H, provided that they are initialised with non-negative values.

As in Standard NMF, the objective is not jointly convex with respect to W and H. We therefore employ an alternating optimisation scheme: the weight matrix $D^{(t)}$ is first recomputed from the current residuals, after which W and H are updated sequentially while the other quantities are held fixed. This procedure is repeated until the stopping criterion is satisfied.

3.2.4 Implementation Procedure

The complete optimisation procedure is summarised as follows:

1. Initialise W and H with non-negative random values.
2. Compute the residual matrix
$$R = X - WH.$$

3. Calculate the sample-wise residual norms

$$\|r_i\|_2 = \|x_i - Wh_i\|_2.$$

4. Construct the diagonal weights using Equation (16).
5. Update W using Equation (18).
6. Update H using Equation (19).
7. Calculate the current objective value

$$J^{(t)} = \sum_{i=1}^{n} \|x_i - W^{(t)} h_i^{(t)}\|_2.$$

8. Repeat Steps 2–7 until the stopping criterion is satisfied or the maximum number of iterations is reached.

Although D is mathematically expressed as an $n \times n$ diagonal matrix, the implementation stores only its diagonal entries as a length-n vector. This avoids explicitly constructing a large diagonal matrix and reduces both memory consumption and computational cost.

3.2.5 Initialisation and Stopping Criterion

The entries of the initial factor matrices $W^{(0)}$ and $H^{(0)}$ were independently sampled from a uniform distribution $\mathcal{U}(0, s)$, where

$$s = \sqrt{\frac{\max\left(\text{mean}(X), \epsilon\right)}{r}}. \tag{20}$$

The experiments were repeated using five random seeds, $\{42, 43, 44, 45, 46\}$. Within each run, Standard NMF and $L_{2,1}$-NMF used identical initial matrices to ensure a fair comparison.

The algorithm terminates when either the maximum number of iterations 500 is reached or the relative change in the objective function satisfies

$$\frac{\left|J^{(t-1)} - J^{(t)}\right|}{\max\left(J^{(t-1)}, \epsilon\right)} < \tau, \tag{21}$$

where τ is the convergence tolerance, set to 10^{-5}.

3.2.6 Robustness to Occlusion Noise

The occlusion noise used in this study introduces contiguous regions with large pixel values into the face images. Under the squared Frobenius objective, images with large reconstruction errors can dominate the optimisation because their residuals are squared. By contrast, the $L_{2,1}$ objective grows linearly with each sample's Euclidean reconstruction error. For example, suppose that two samples have residual magnitudes a and ca, where $c > 1$. Under the squared Frobenius loss, the second sample contributes c^2 times as much loss as the first sample. Under the $L_{2,1}$ loss, it contributes only c times as much. Therefore, the relative influence of a heavily corrupted sample grows less rapidly under the $L_{2,1}$ objective.

Furthermore, the reweighting mechanism in Equation (16) automatically assigns smaller weights to images with larger residuals. Consequently, severely corrupted images have a reduced effect on the learned basis matrix W. We therefore expect $L_{2,1}$-NMF to produce a lower relative reconstruction error than Standard NMF as the size or number of occlusion blocks increases.

However, $L_{2,1}$-NMF performs robustness weighting at the sample level rather than directly identifying individual corrupted pixels. Therefore, its advantage may be less pronounced when every image contains a similar amount of occlusion. This limitation is examined empirically by varying the fraction of corrupted images in the dataset.

3.2.7 Computational Complexity

The main computational cost in each iteration arises from matrix multiplications used to update W and H. For $X \in \mathbb{R}^{m \times n}$ and factorisation rank r, the approximate per-iteration complexity is

$$\mathcal{O}(mnr), \tag{22}$$

which is of the same leading order as Standard NMF. The additional cost of $L_{2,1}$-NMF comes from calculating the residual norms and sample weights at each iteration.

3.3 Occlusion Noise Generation

We evaluate the robustness of the NMF algorithms using contiguous occlusion noise. This type of corruption modifies pixels in spatially adjacent regions and simulates partial facial occlusions, such as objects covering the eyes, nose, or mouth.

All images were converted to grayscale, and their pixel intensities were normalised from $[0, 255]$ to $[0, 1]$. The ORL images were resized from 92×112 to 30×37 pixels using a reduction factor of 3, while the Extended YaleB images were resized from 168×192 to 42×48 pixels using a reduction factor of 4.

Let $\widehat{X}$ denote the clean data matrix and let X denote the corrupted data matrix used as input to the NMF algorithms. For each image, q square blocks with an edge length of b pixels were placed at randomly selected locations. All pixels within each block were replaced with the maximum normalised pixel intensity of 1. The block locations were sampled independently for each image while ensuring that every block remained within the image boundary.

We varied both the block edge length b and the number of blocks q to produce different levels of corruption. The experimental settings are summarised in Table 3.

Table 3: Occlusion noise settings used for the ORL and Extended YaleB datasets.

Dataset	Experiment	Block size b	Blocks q
ORL	Clean	0	0
ORL	Block size	$\{3, 6, 9\}$	1
ORL	Block number	6	$\{1, 2, 3\}$
Extended YaleB	Clean	0	0
Extended YaleB	Block size	$\{4, 8, 12\}$	1
Extended YaleB	Block number	8	$\{1, 2, 3\}$

4 Experiments

4.1 Evaluation Metrics

For this section, let V denote the contaminated dataset (by adding noise), and $\hat{V}$ denote the clean dataset. Let W and H denote the factorization results on V. Finally, let Y denote the true labels and Y_{pred} denote the predicted labels obtained from the reconstructed data.

4.1.1 Relative Reconstruction Errors (RRE)

The RRE measures the ratio of the Frobenius norm of the residuals to the Frobenius norm of the clean dataset, and will be the main metric used to compare the quality of reconstructions.

$$RRE = \frac{\|\hat{V} - WH\|_F}{\|\hat{V}\|_F}$$

A drawback of RRE is that it is biased towards the Standard NMF algorithm, as they both share a Frobenius norm minimizer. On the other hand, a single $L_{2,1}$ norm value may correspond to multiple RRE values (triangle inequality), so minimizing one may not always minimize the other.

4.1.2 Average Accuracy

To evaluate the accuracy of predictions made on reconstructed data, we first use scikit-learn's KMeans clustering algorithm to create clusters equal to the number of label, then assign each cluster the majority true label among its members as Y_{pred} (note that clusters may share the same label).

$$Acc(Y, Y_{pred}) = \frac{1}{n} \sum_{i=1}^{n} 1\{Y_{pred}(i) == Y(i)\}$$

4.1.3 Normalized Mutual Information (NMI)

Using Y_{pred}, we can also calculate the NMI using mutual information $I(.,.)$ and entropy $H(.)$.

$$NMI(Y, Y_{pred}) = \frac{2 * I(Y, Y_{pred})}{H(Y) + H(Y_{pred})}$$

Both average accuracy and NMI can be rather unstable as they compound reconstruction error with prediction error, which depends on the dataset itself. Clustering on the original datasets demonstrate that accuracy and NMI are not suitable metrics for the Cropped YaleB dataset.

Table 4: Clustering based evaluation metrics on original dataset

Dataset	ACC (mean)	NMI (mean)
ORL	0.7400	0.8603
Cropped YaleB	0.1139	0.1215

4.2 Experimental Setup

For the following experiments, we will refer to the following parameters:

- rank refers to the factorization rank, default = number of label classes
- block_size refers to the length of occlusion blocks (in pixels), default = 1
- num_blocks refers to the number of occlusion blocks generated per image, default = 100
- fraction refers to the fraction of corrupted images in the dataset, default = 0.1

Additionally, each run will be capped at a maximum of 1500 optimization iterations with an early stoppage tolerance of 1e-5, unless otherwise specified. For optimization purposes, we will also use an image downsampling factor of 3 for the ORL dataset, and 5 for the Cropped YaleB dataset throughout.

To mitigate experimental errors as much as practically possible, each experiment will be run 5 times, with results averaged for reporting. Each time, a different 80% slice of the dataset will be used, unless otherwise specified. A different seed will also be used to select images to corrupt, initialize W and H for models, and perform KMeans clustering. The seeds used are: [4, 5, 3, 2, 8]

4.3 Demonstration of Outliers

This preliminary test demonstrates how our occlusion noise injection methods simulate outliers. A train-test-split of 90%-10% was used on the ORL dataset, and the standard model was used to obtain W from clean training data, which is then fitted to a clean dataset and a fully corrupted dataset, both derived from the same test data. Results demonstrate a deviation in RRE around one magnitude larger than either standard deviation.

Table 5: Performance of basis learned from clean data on unseen data

Unseen Data	RRE (mean)	RRE (std)
Clean	0.1236	0.005574
Corrupt	0.1713	0.007004

4.4 Reconstruction of Single Dataset

Experiments in this section examine both models' ability to reconstruct clean data from a single corrupted dataset. The ORL dataset will be used here as it is better suited for clustering metrics.

4.4.1 Factorization Rank

This experiment trains both models on a clean dataset, with varying rank values: [20, 30, 40, 50, 60, 70, 80, 90]. Results demonstrate that both models produce generally similar results, with a near-identical RRE curve. The RRE curve is convex and decreases monotonically as rank increases. On the other hand, accuracy and NMI values increase to a maximum at around rank = 40, before decreasing as rank further increases.

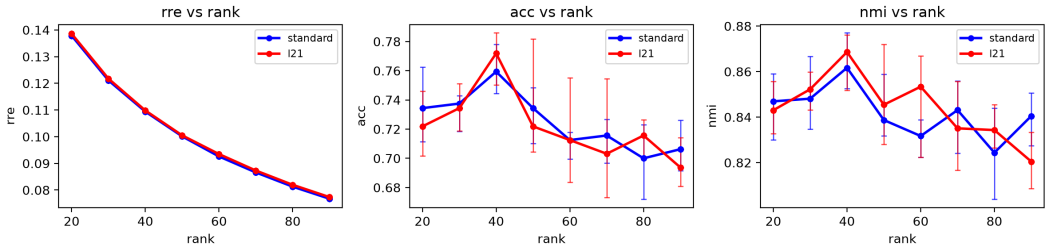


Figure 1: Metrics vs Factorization Rank (fraction = 0)

This experiment is similar to the previous, but uses a corrupted dataset with default values for training instead. Results demonstrate a convex RRE curve for the $L_{2,1}$ model that hits a minimum near rank = 70, whereas the standard RRE curve decreases initially, then rises to around rank = 60 and falls again. The two curves diverge near the start and begin to converge near the end. The accuracy and NMI values of both models are generally decreasing, with the standard model decreasing at a faster rate.

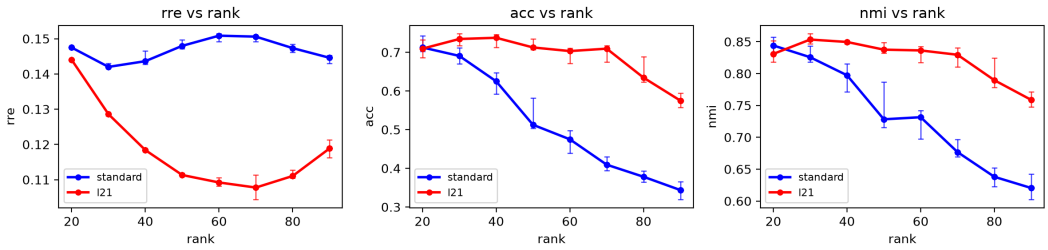


Figure 2: Metrics vs Factorization Rank (fraction = 0.1)

4.4.2 Number of Occlusion Blocks

This experiment trains both models on a corrupted dataset with increasing num_blocks values: [0, 50, 100, 150, 200, 250, 300, 350]. Results demonstrate two concave RRE curves that are monotonically increasing, with the $L_{2,1}$ curve increasing at a slower rate. The accuracy and NMI values are generally decreasing, with the $L_{2,1}$ model decreasing at a slower rate.

Reconstruction comparisons on the same corrupted sample also show a clear, accurate reconstruction of the corrupted data for the standard model, whereas the $L_{2,1}$ reconstruction is blurred with no visible occlusion noise.

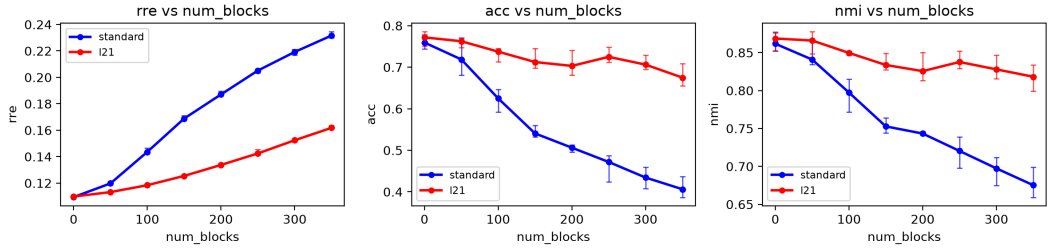
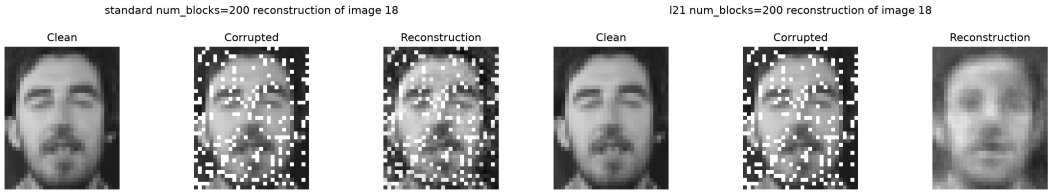


Figure 3: Metrics vs Number of Occlusion Blocks



(a) Reconstruction by standard NMF.

(b) Reconstruction by $L_{2,1}$-NMF.

Figure 4: Reconstruction of the same sample under severe corruption with 200 occlusion blocks.

4.4.3 Size of Occlusion Blocks

This experiment trains both models on a corrupted dataset with increasing block_size values: [0, 2, 4, 6, 8, 10, 12, 14]. Results demonstrate two monotonically increasing convex RRE curves, with the $L_{2,1}$ curve just slightly below the standard. The accuracy and NMI values are also generally decreasing, with the $L_{2,1}$ curve slightly above the standard.

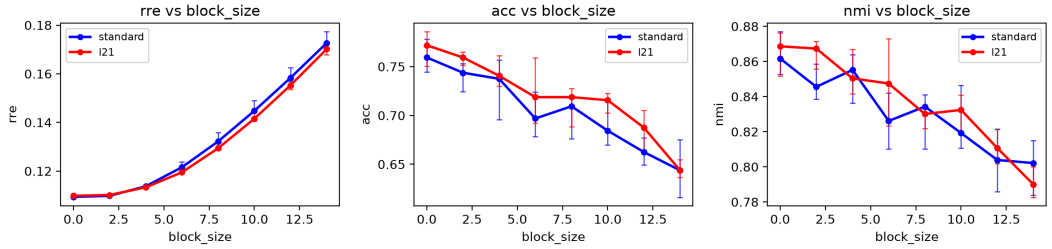


Figure 5: Metrics vs Size of Occlusion Blocks

4.4.4 Same Degree of Corruption

This experiment trains both models on a corrupted dataset with (block_size, num_blocks) values: [(1, 256), (2, 64), (4, 16), (8, 4), (16, 1)] while keeping the total number of occlusion pixels per corrupted image at 256. Results demonstrate a monotonically increasing $L_{2,1}$ RRE curve as block_size increases, while the standard counterpart is monotonically decreasing. Both curves converge toward increasing block_size. The accuracy and NMI values demonstrate a similar pattern, but with the $L_{2,1}$ above and the standard below.

Reconstruction comparisons for the $L_{2,1}$ model on a corrupted sample shows no occlusion noise explicitly captured when num_blocks = 256, but a visible patch of occlusion noise captured when

block_size = 16. In contrast, comparisons for the standard model on a clean sample shows a better reconstruction at block_size = 16 compared to num_blocks = 256.

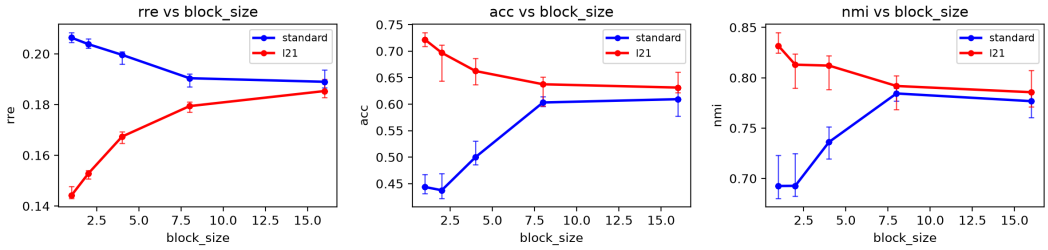


Figure 6: Metrics vs block_size (at 256 noise pixels total)

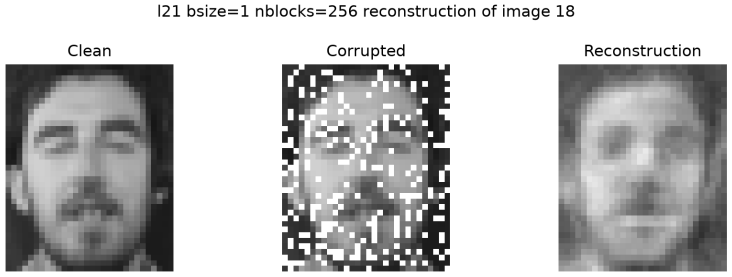


Figure 7: $L_{2,1}$ Reconstruction when block_size = 1, num_blocks = 256

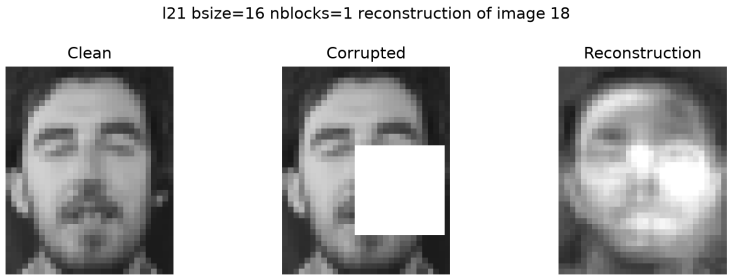


Figure 8: $L_{2,1}$ Reconstruction when block_size = 16, num_blocks = 1

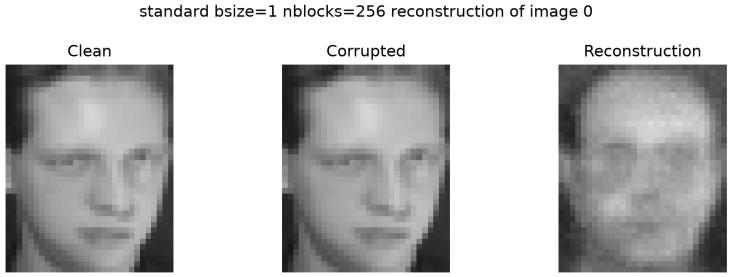


Figure 9: Standard Reconstruction when block_size = 1, num_blocks = 256



Figure 10: Standard Reconstruction when block_size = 16, num_blocks = 1

4.4.5 Fraction of Corrupted Images

This experiment trains both models on a training set with varying fractions of corrupted images: [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]. Results demonstrate a standard RRE curve that is concave in shape, increasing sharply at first, then gradually; whereas the $L_{2,1}$ curve below it is roughly linear. Both curves meet at fraction = 0.0 and fraction = 1.0. The accuracy and NMI lines of both models meet at the same two points, with the $L_{2,1}$ curve consistently above the standard in between.

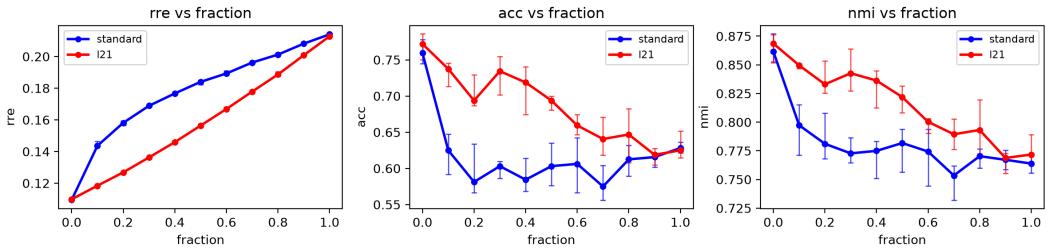


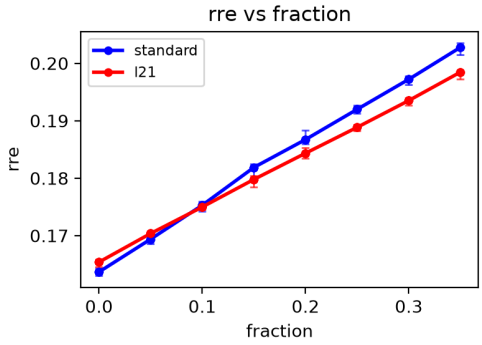
Figure 11: Metrics vs Corrupted Fraction

4.5 Reconstruction of Unseen Dataset

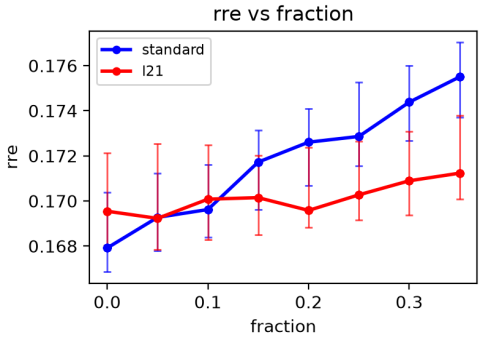
Experiments in this section will use a train-test split of 70%-30% from the larger Cropped YaleB dataset. The basis W learned from the training data will then be fixed when fitting H to the test data, and the final reconstructions will be evaluated using the clean test data.

4.5.1 Corrupt to Clean

This experiment trains both models on a training set with varying fractions of corrupted images: [0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35]. The learned W matrix under each setting will then be used to fit the clean test data. Results demonstrate a seemingly linear increase in RRE of both models as fraction increases, when evaluated on both the clean training data and clean test data. The overall RRE values of both models rise more slowly on the test set, and the narrower scale reveals that the $L_{2,1}$ RRE values are higher at and before fraction = 0.1, but grows at a slower rate and falls below the standard beyond that point.



(a) Evaluation on the clean training set.



(b) Evaluation on the test set.

Figure 12: Evaluation results for corrupt-to-clean reconstruction on the training and test sets.

4.5.2 Clean to Corrupt

This experiment trains both models on a clean training set, then uses the learned W matrix to fit a test set with varying fractions of corrupted images: [0.0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35]. Results demonstrate a seemingly linear increase in RRE of both models as fraction increases, with both lines close together but the $L_{2,1}$ slightly above.

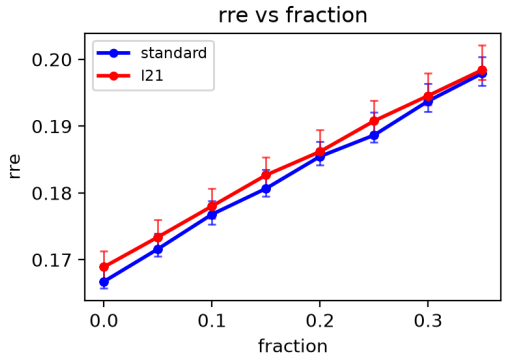


Figure 13: Evaluation on the test set (clean to corrupt)

4.6 Discussion

We primarily use the relative reconstruction error (RRE) in this discussion, because it varies less across runs than the clustering metrics, accuracy (ACC) and normalized mutual information (NMI), whose standard deviations are much larger when only a small number of runs is available. This is visible in the plots. RRE is based on the Frobenius norm and therefore slightly favours the standard model, as discussed in Section 4.1, so we note where this may affect a comparison. When comparing the two models under the same setting, the metrics generally agree: a model with consistently lower RRE tends to have higher ACC and NMI on average, and larger differences in RRE tend to accompany larger differences in ACC and NMI. Cases where the clustering metrics disagree with RRE are examined separately.

Section 3.2.6 describes the robustness of the $L_{2,1}$ model to sample-level outliers [5]. In the context of facial images, a sample-level outlier is an image whose features differ markedly from the rest of the dataset, and which therefore has a disproportionately large reconstruction error under components learned from the other samples. To simulate such outliers in a controlled way, we corrupt a fraction of the images with randomly placed white blocks, varying their number and size, which disrupts the facial structure that the model is trying to learn. Section 4.3 demonstrates the effect of this corruption on the RRE.

4.6.1 Magnitude of Outliers

Sections 3.1.6 and 3.2.6 describe how the two objectives treat sample-level outliers. Under the squared Frobenius norm, a sample's loss grows quadratically with its residual norm and its gradient grows linearly, so a poorly fitted outlier exerts ever more influence and pushes the model to fit it at the expense of clean samples. Under the $L_{2,1}$ norm, the loss grows only linearly and the gradient with respect to the residual has constant unit norm, so every sample pulls on the solution with equal strength regardless of how badly it fits [5]. This bounds the influence of outliers but does not eliminate it; because the $L_{2,1}$ loss remains unbounded, errors far from zero can still skew the objective, motivating truncated loss functions that cap large errors entirely [8].

This is directly observed in Fig. 3, where the performance of the $L_{2,1}$-NMF model degrades more slowly and consistently than that of standard NMF as the severity of sample-level outliers increases. This indicates that $L_{2,1}$-NMF is less affected by heavily corrupted samples. The reconstruction comparison in Fig. 4 further supports this result. Under severe corruption with 200 occlusion blocks, standard NMF reproduces the corrupted pixels and their surrounding structure (Fig. 4a), whereas $L_{2,1}$-NMF largely avoids reproducing the corruption (Fig. 4b).

Interestingly, this advantage largely disappears in Fig. 5, where the severity of sample-level outliers is increased through structured square blocks instead of isolated random pixels. The losses alone do not explain this: each sample's contribution depends only on its residual, and the two corruption types affect the same number of pixels. The difference must therefore lie in how well the shared components can represent the corruption, which determines the residual itself.

Figures 7, 8, 9 and 10 suggest a plausible explanation. Isolated random pixels fall at different locations in different images and share little structure, so fitting them requires sacrificing components that would otherwise capture clean samples. The standard model's quadratic loss forces this trade-off, while the $L_{2,1}$ model down-weights the affected samples. Contiguous blocks concentrate the corruption in overlapping image regions, so the corrupted samples resemble one another and a few shared components can absorb the corruption at little cost to the clean samples. Both models can then accommodate it, which explains the convergence in Fig. 6, where merging random pixels into blocks makes the corruption more overlapping and learnable.

Hence, although the $L_{2,1}$ norm limits the influence of sample-level outliers, its benefit depends on the structure of the corruption and its relationship to the rest of the dataset. Since NMF compresses the data into shared components, a corruption that is itself low-rank across samples is hard to distinguish from useful information [15].

4.6.2 Factorization Rank

Increasing the factorization rank reduces the compression applied to the data and allows the basis to store more information, which improves reconstruction on a fixed dataset, as seen in Fig. 1. The gains are largest at low ranks and diminish thereafter, consistent with the first components capturing the most widely shared structure and later ones refining increasingly specific details. Exact reconstruction is possible once the factorization rank reaches the nonnegative rank of the data matrix; in the worst case this can be as large as the number of samples.

Interestingly, accuracy and NMI do not follow this trend. In Fig. 1, both are highest when close to the number of classes, and decline as the rank increases beyond. A plausible explanation is that when the rank is close to the number of classes, each component can specialise in a single class, so the coefficient vectors approximate cluster indicators and suit K-Means [1]. Additional components are more likely to capture variation shared across classes, such as illumination, which further reduces the RRE but makes the representations of different classes more similar and degrades clustering.

When the training data contains sample-level outliers, additional capacity can also be spent on fitting the corruption rather than the clean structure. As discussed in Sections 3.1.6 and 3.2.6, the squared Frobenius loss amplifies large residuals, so the standard model is particularly prone to this, whereas the $L_{2,1}$ model limits the influence of the corrupted samples [5]. Fig. 2 reflects this: as the rank increases, the performance of the standard model degrades, while that of the $L_{2,1}$ model improves. The slight initial improvement of the standard model suggests that at the lowest ranks the components still mostly capture structure shared by the clean samples, whose combined residual outweighs that of the outliers. As the rank grows, the standard model increasingly dedicates components to re-

producing the outliers, whereas the $L_{2,1}$ model dedicates them to the clean structure. As the rank approaches the number of samples, both models can reproduce every sample, corrupted or not, so their performance converges.

Therefore, in the presence of sample-level outliers, a higher rank does not necessarily give better performance. The rank must therefore be selected explicitly; rank-selection procedures based on clustering stability have been proposed for NMF-based clustering [16]. The robustness of the $L_{2,1}$ model nonetheless allows it to reach a better optimum than the standard model.

4.6.3 Proportion of Outliers

Besides the magnitude of sample-level outliers, their proportion in the dataset also affects training. Both objectives sum the sample-level contributions to the loss, so what matters is the share of the total loss attributable to outliers. When outliers are sparse, converting one more clean sample into an outlier changes this share substantially. Because the squared Frobenius norm amplifies large residuals quadratically, this share is larger in the standard model than in the $L_{2,1}$ model when outliers are sparse, which forces the standard model to devote more of its capacity to fitting the noise. This accounts for the divergence between the two models at low corruption fractions in Fig. 11.

As the corrupted fraction grows, the outliers' share of the loss approaches one in both models, and each additional outlier of the same magnitude changes it less. This saturation is faster under the squared Frobenius norm, since quadratic amplification lets the outliers dominate the loss earlier, whereas the linear growth of the $L_{2,1}$ loss approaches it more slowly. The additional outliers therefore matter progressively less to both objectives.

When most samples are similarly corrupted, their residuals are of comparable size, so each sample receives approximately the same weight in both models. The two models then learn a similar representation of the data, and their performance converges to a similar, and poor, level, as seen in Fig. 11. Intuitively, once corrupted samples form the majority, the dataset no longer contains outliers but a complete distribution shift, and the notion of down-weighting atypical samples loses its meaning.

This supports the claim that the $L_{2,1}$ model is robust to sample-level outliers, but it also shows that its reweighting mechanism makes little difference under dataset-wide noise. Such noise is better handled by a loss matched to the noise distribution, such as L_1-type losses for Laplacian noise [4], which we leave to future work.

4.6.4 Generalization Ability

When both models are trained on data containing sample-level outliers, we expect the $L_{2,1}$ model to learn a basis that generalizes better to unseen clean data. In the update step for W in (18), each sample's contribution is weighted by the matrix D according to its residual, which limits the influence of outliers on the basis, whereas in the standard model their influence grows with the size of their residual.

To test this, we train both models on data with an increasing fraction of outliers and evaluate the reconstruction of held-out clean samples, with H fitted on the test data using the learned basis. As a sanity check, Figs. 12a and 12b present the corrupt-to-clean reconstruction results on the training and test sets, respectively. As shown in Fig. 12a, the training reconstructions produced by the $L_{2,1}$-NMF model degrade more slowly as the proportion of corrupted samples increases. A similar trend is observed on the test set in Fig. 12b, indicating that the basis learned by $L_{2,1}$-NMF is more stable and generalises better to unseen clean samples than that learned by standard NMF.

Comparing the two figures, the test RRE rises more slowly than the training RRE. This is expected: the training images include the corrupted samples, which are intrinsically harder to reconstruct, whereas the test images are clean, so their RRE is primarily affected through the degradation of the learned basis. The relative standard deviation is also higher on the test set. This is likely due to its smaller size, which increases sampling variance, and possibly to differences in split quality. At low corruption fractions, the standard model is marginally better than the $L_{2,1}$ model. This may be attributed to the bias of the RRE metric, which is based on the Frobenius norm and therefore favours the model that optimizes it directly (Section 4.1).

These results show that the advantage of the $L_{2,1}$ algorithm extends beyond reconstructing the training data, to reconstructing unseen clean samples from the learned basis. This suggests the basis should also be more reliable for downstream uses such as feature extraction when the training data contains outliers, although we do not test this directly.

4.6.5 Test Time Corruption

Conversely, when both models are trained on clean data and then fitted to corrupted test data (Section 4.5.2), their performance remains similar as the corruption fraction increases. The update steps for H in (5) and (19) are identical: D is applied column-wise to both numerator and denominator and cancels in the element-wise ratio. With W fixed, each column of H solves an independent problem, and the $L_{2,1}$ objective for a column is a monotone function of the squared error minimized by the standard model, so both models have the same minimizer. Since the two bases are of similar quality when trained on clean data, both models achieve similar performance on the test data.

This shows that the advantage of the $L_{2,1}$ algorithm does not extend to distribution shifts at test time. The reweighting applies across samples through D, but the representation of each test sample is fitted independently, so it cannot learn to ignore noise during fitting. The robustness of the $L_{2,1}$ model is therefore tied to its learned basis, and handling test-time corruption would require a different encoding step, such as an entry-wise robust loss on H or an explicit sparse error term.

5 Conclusion

Our experiments have demonstrated that the $L_{2,1}$ norm is better suited for mitigating the effects of extreme outliers in datasets, compared to the squared Frobenius norm. Its robustness comes from its magnitude-based reweighting mechanism, which helps it to learn components that are less biased towards outliers and outperform the standard variant when outliers are unavoidable in training.

One caveat to keep in mind is the nature of outliers. It is completely possible for outliers to be present within clean datasets and carry meaningful information. (For example, underrepresented minorities in a facial dataset) In such a case, using an outlier-robust algorithm would hurt more than help, and a better solution would be to increase the sampling size. Alternatively, structural biases might cause inputs to deviate similarly across training, testing and deployment. In such a case, learning the noise might prove more useful for classification or regression if the underlying bias cannot be removed.

The $L_{2,1}$ norm targets outliers with large residuals, so its advantage is limited when noise is spread across every sample. Several directions could address this:

- Using alternative loss functions, such as the L_1 norm which penalises entry-wise residuals and suits sparse pixel-level corruption such as salt-and-pepper noise.
- Denoising strategies, such as PCA denoising, which can help identify noise components as high variance components.
- Alternative models, such as denoising autoencoders, which can be trained to produce compressed representations that can be reconstructed, removing input noise in the process.

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